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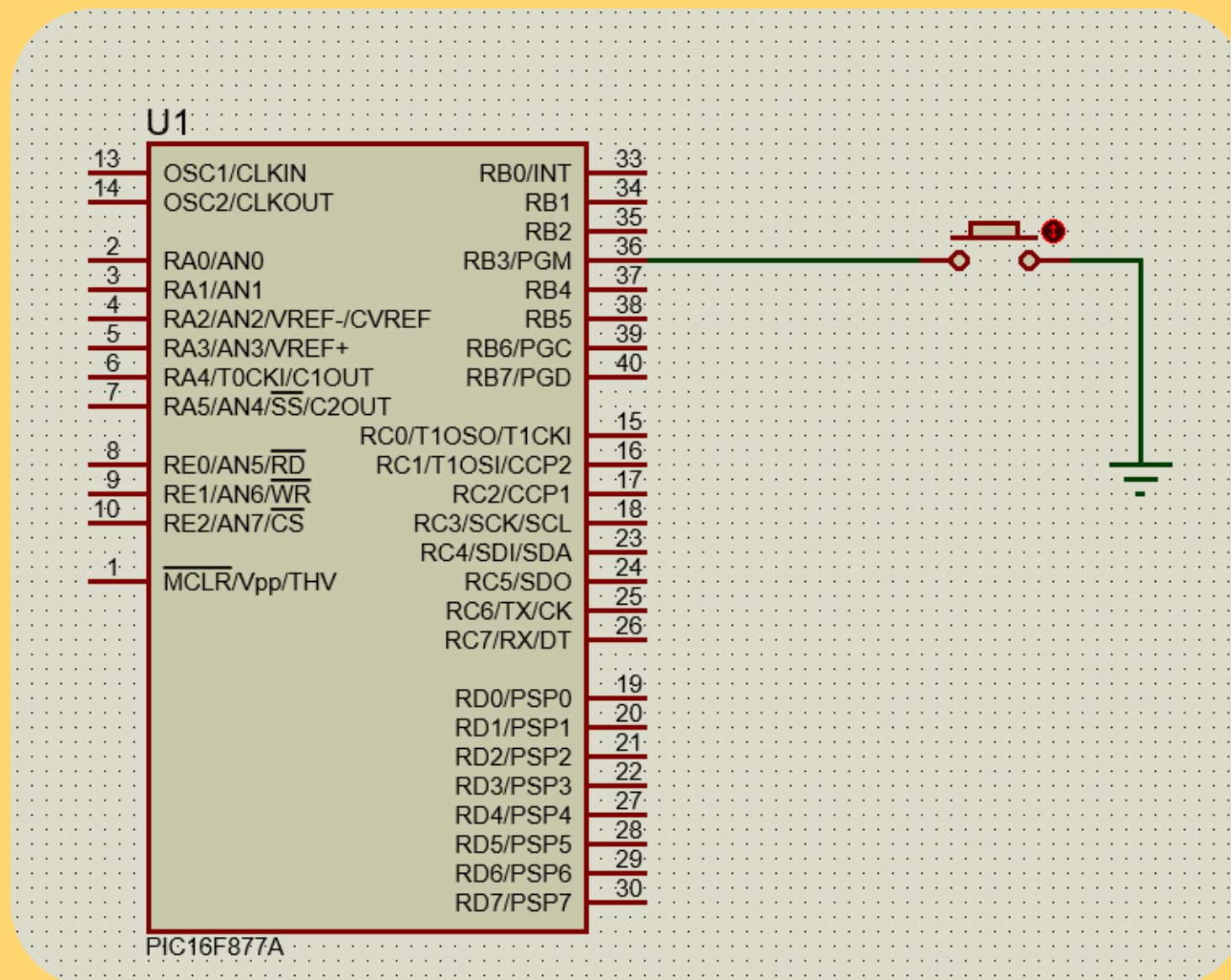
How to Interface a Switch

Pull-UP!

”



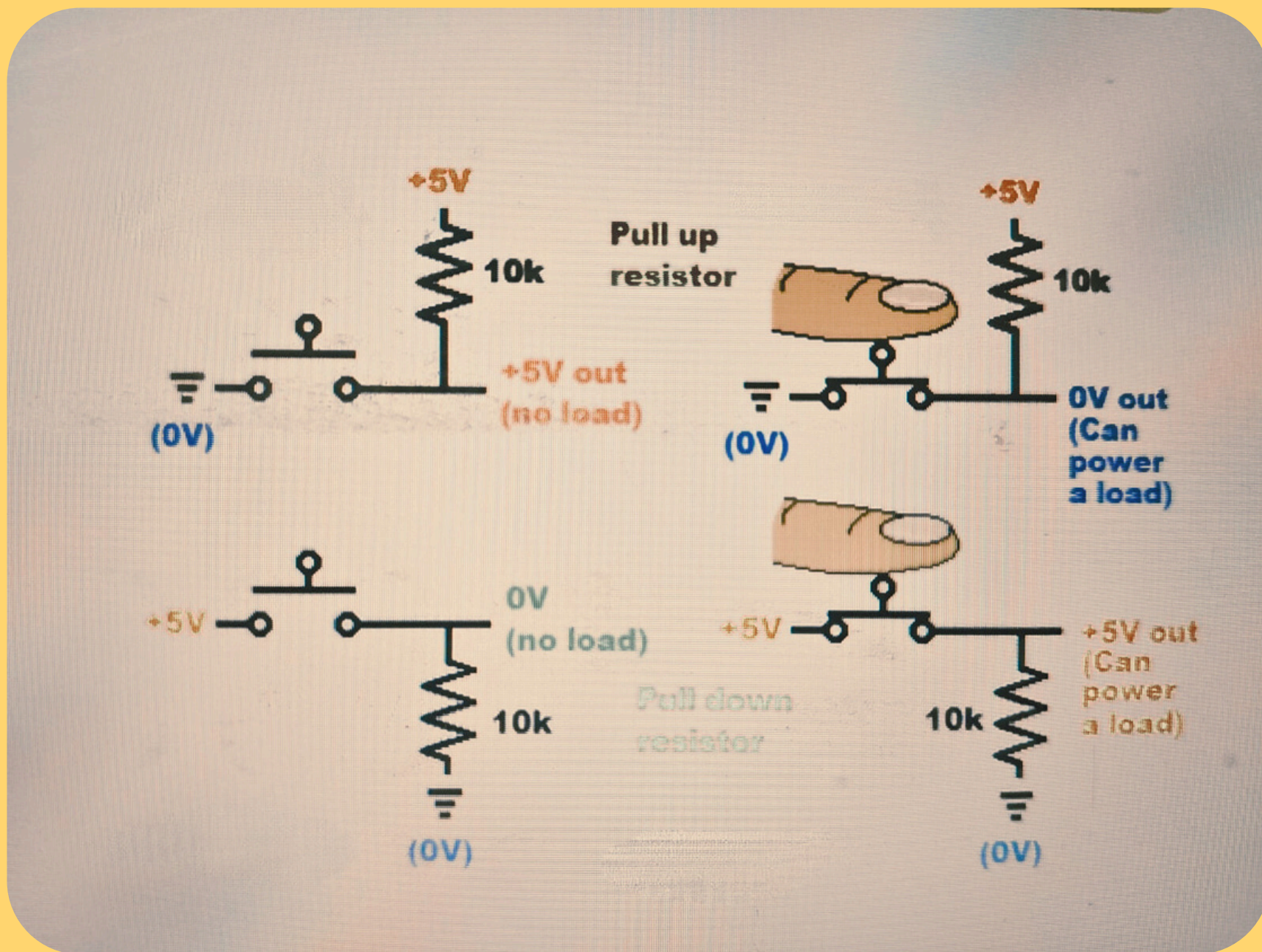
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If we configure RB3 as an input pin,

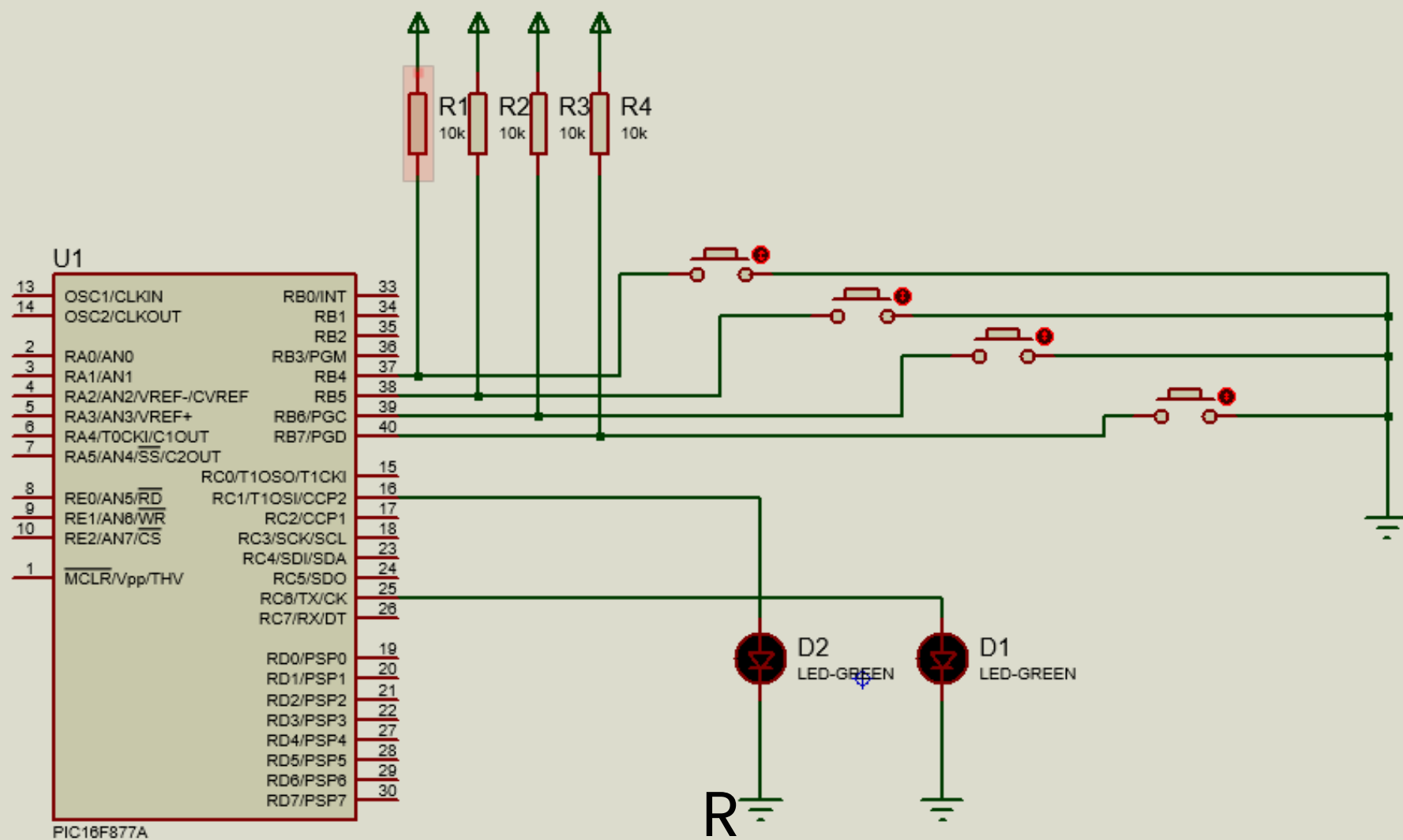
- When the button is pressed, the pin will be at logic 0 (LOW).
- When the button is not pressed, the pin becomes floating, meaning its state can randomly fluctuate between 0 and 1.

To avoid this unstable behavior, we need to fix the pin's state when the button is not pressed — this is done using a pull-up or pull-down resistor.



- When a **pull-up** resistor is connected, the pin stays at a **HIGH** state when the switch is not pressed.
- When a **pull-down** resistor is connected, the pin stays at a **LOW** state when the switch is not pressed.



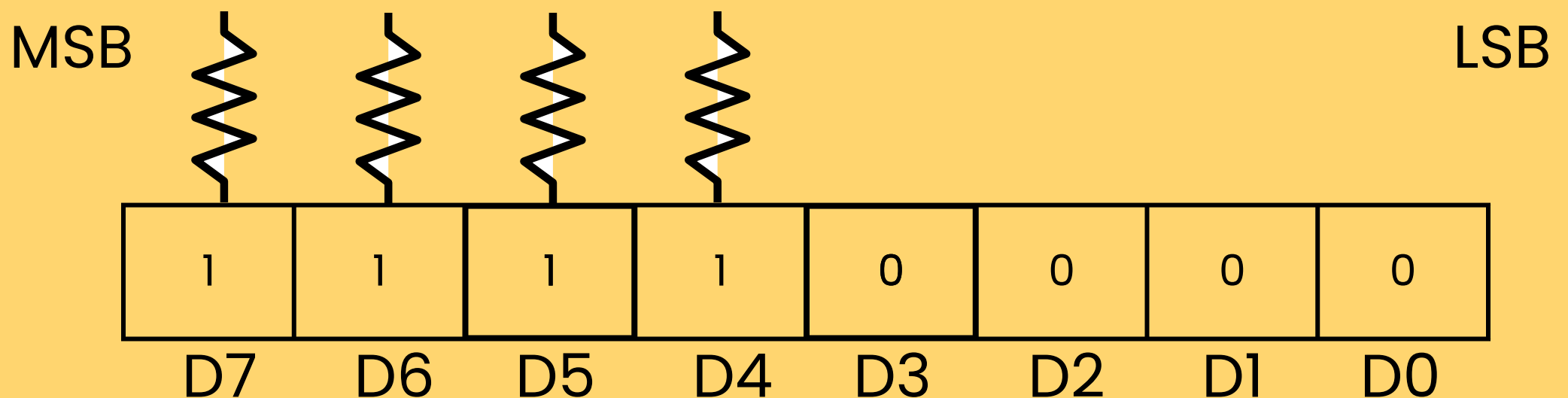


Goal: Connect switches with pull-up configuration and create an LED blinking pattern.

	RC1	RC6
RB4	1	0
RB5	1	1
RB6	0	1
RB7	0	0



We need to identify the PORTB value, compare it, and glow the LED accordingly.



If no button is pressed, PORTB reads a **default** value of **0xF0** due to the pull-up configuration.

If **RB4** is pressed PORTB = **0xE0**

1	1	1	0	0	0	0	0
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If **RB5** is pressed PORTB = **0xD0**

1	1	0	1	0	0	0	0
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Similarly its 0xB0, 0x70 when RB6 , RB7 pressed.

```

void main(void) {

    TRISB = 0XF0; // 1111 0000 Making RB(4-7) pins as Input
    TRISC = 0X00; // 0000 0000 Making All pins as Input

    PORTC = 0X00; // 0000 0000 Making All pins to low state
    PORTB = 0X00; // 0000 0000 Making All pins to low state

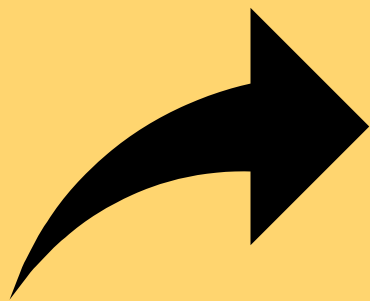
    while(1){
        value = PORTB; // Reading the PORTB registers value
        switch(value){
            case 0xE0: // 1110 0000 When button connected on RB4 pressed
                PORTC = 0x02; // 0000 0010
                break;
            case 0xD0: // 1101 0000 When button connected on RB5 pressed
                PORTC = 0x42; // 0100 0010
                break;
            case 0xB0: // 1011 0000 When button connected on RB6 pressed
                PORTC = 0x40; // 0000 0010
                break;
            case 0x70: // 0111 0000 When button connected on RB7 pressed
                PORTC = 0x00; // 0000 0000
                break;
            default : // When button is released
                PORTC = 0x00; // 0000 0000
                break;
        }
    }
    return;
}

```

Depending on the switch pressed, the LEDs connected to PORTC will glow accordingly.



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